

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.2467848571904588, median 0.23563842426067005, std: 0.12558162292924097
Reprojection error (cam1): mean 0.22165632382677544, median 0.2055593717475851, std: 0.12256201930234488
Gyroscope error (imu0): mean 0.28123328316922186, median 0.2272668208657208, std: 0.23055125919708602
Accelerometer error (imu0): mean 0.4316779981441826, median 0.4189656042041089, std: 0.17886298047923685

Residuals

Reprojection error (cam0) [px]: mean 0.2467848571904588, median 0.23563842426067005, std: 0.12558162292924097
Reprojection error (cam1) [px]: mean 0.22165632382677544, median 0.2055593717475851, std: 0.12256201930234488
Gyroscope error (imu0) [rad/s]: mean 0.006288567387018275, median 0.005081840604860193, std: 0.005155282878628579
Accelerometer error (imu0) [m/s^2]: mean 0.09652613482414206, median 0.09368355712346592, std: 0.03999497830097915

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[ [ 0.01120139  0.99992559 -0.00483191  0.07438326]
  [ 0.99988601 -0.01124961 -0.01007049 -0.00222439]
  [-0.01012409 -0.00471856 -0.99993762 -0.0087888 ]
  [ 0.          0.          0.          1.          ] ]
```

T_ic: (cam0 to imu0):

```
[ [ 0.01120139  0.99988601 -0.01012409  0.00130197]
  [ 0.99992559 -0.01124961 -0.00471856 -0.07444422]
  [-0.00483191 -0.01007049 -0.99993762 -0.00845124]
  [ 0.          0.          0.          1.          ] ]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
0.00180669212067735

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.00091935 0.99999205 -0.00388034 -0.00090486]
[0.99994187 -0.00096097 -0.01073902 -0.00025911]
[-0.01074267 -0.00387024 -0.99993481 -0.00974467]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[0.00091935 0.99994187 -0.01074267 0.00015525]
[0.99999205 -0.00096097 -0.00387024 0.00086689]
[-0.00388034 -0.01073902 -0.99993481 -0.00975032]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.001985086644975165

Baselines:

Baseline (cam0 to cam1):
[[0.99994668 -0.0102912 -0.00084773 -0.0753145]
[0.01029173 0.99994685 0.00061938 0.00120507]
[0.00084131 -0.00062807 0.99999945 -0.00101985]
[0. 0. 0. 1.]]

baseline norm: 0.07533104517898724 [m]

Gravity vector in target coords: [m/s²]
[0.14186023 -9.80533303 0.06117802]

Calibration configuration

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cam0

Camera model: pinhole
Focal length: [406.22491143638547, 406.4836240690851]
Principal point: [326.28445665863444, 205.63422758496844]
Distortion model: radtan
Distortion coefficients: [0.02707791337649835, -0.08666439949615883, 0.0019570847381826665, 0.000786597327492806]
Type: aprilgrid
Tags:
Rows: 5
Cols: 5
Size: 0.04 [m]
Spacing 0.012 [m]

cam1

Camera model: pinhole
Focal length: [402.2899512482709, 402.5453596457853]
Principal point: [305.45614012706596, 179.2919040597012]
Distortion model: radtan
Distortion coefficients: [0.04035621532991146, -0.11543545682000089, 0.0007003772155267906, 0.0009699249340991044]
Type: aprilgrid
Tags:
Rows: 5
Cols: 5
Size: 0.04 [m]
Spacing 0.012 [m]

IMU configuration

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IMU0:

Model: calibrated
Update rate: 125.0

Accelerometer:

Noise density: 0.02

Noise density (discrete): 0.22360679774997896

Random walk: 0.0002

Gyroscope:

Noise density: 0.002

Noise density (discrete): 0.022360679774997897

Random walk: 2e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

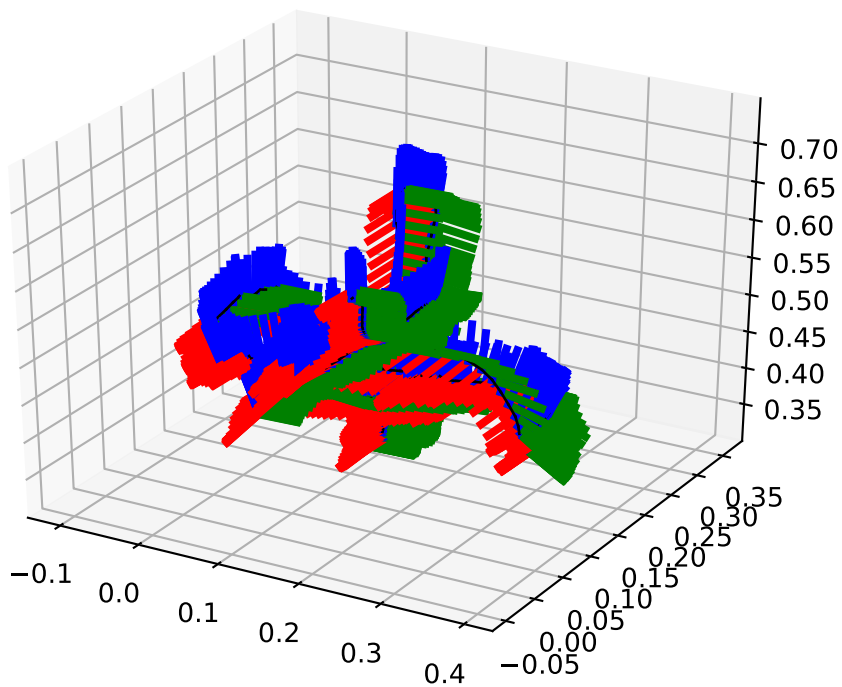
[0. 1. 0. 0.]

[0. 0. 1. 0.]

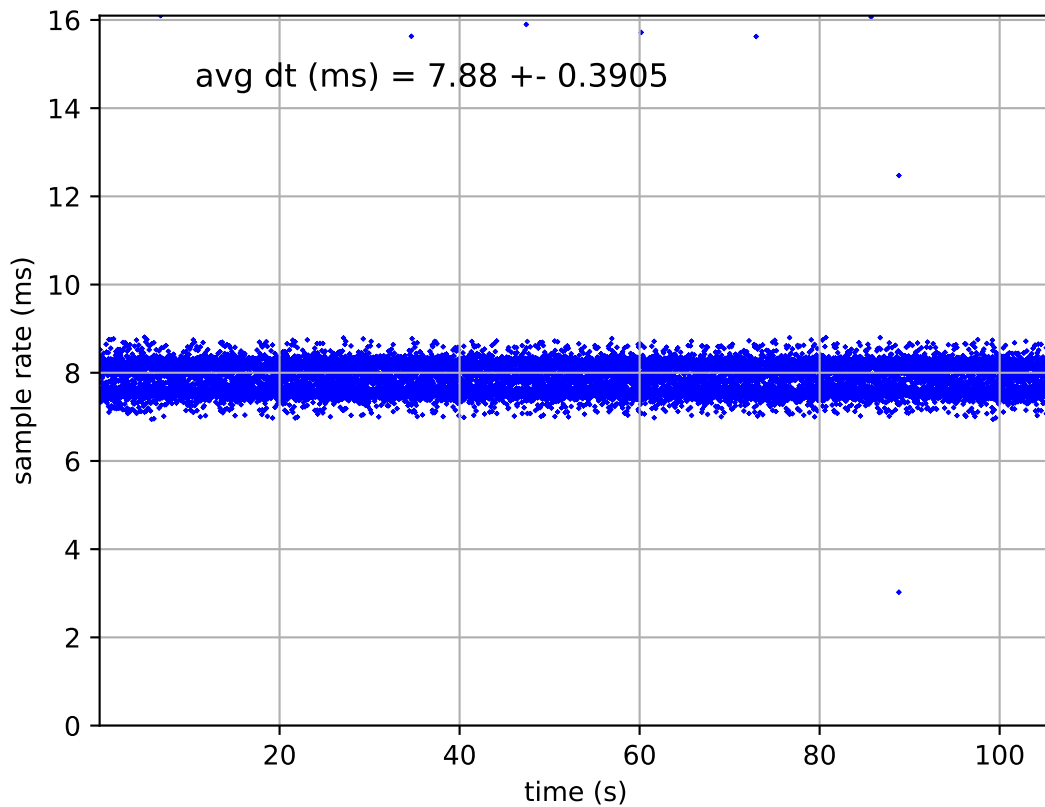
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

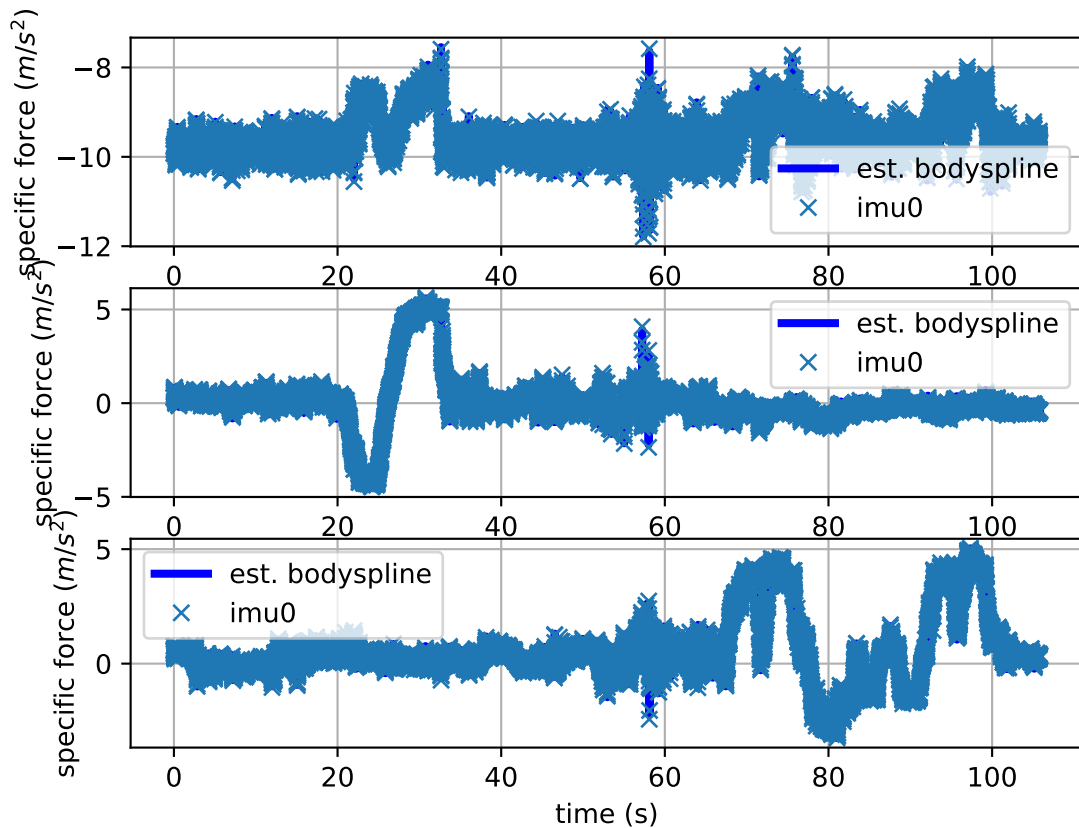
imu0: estimated poses



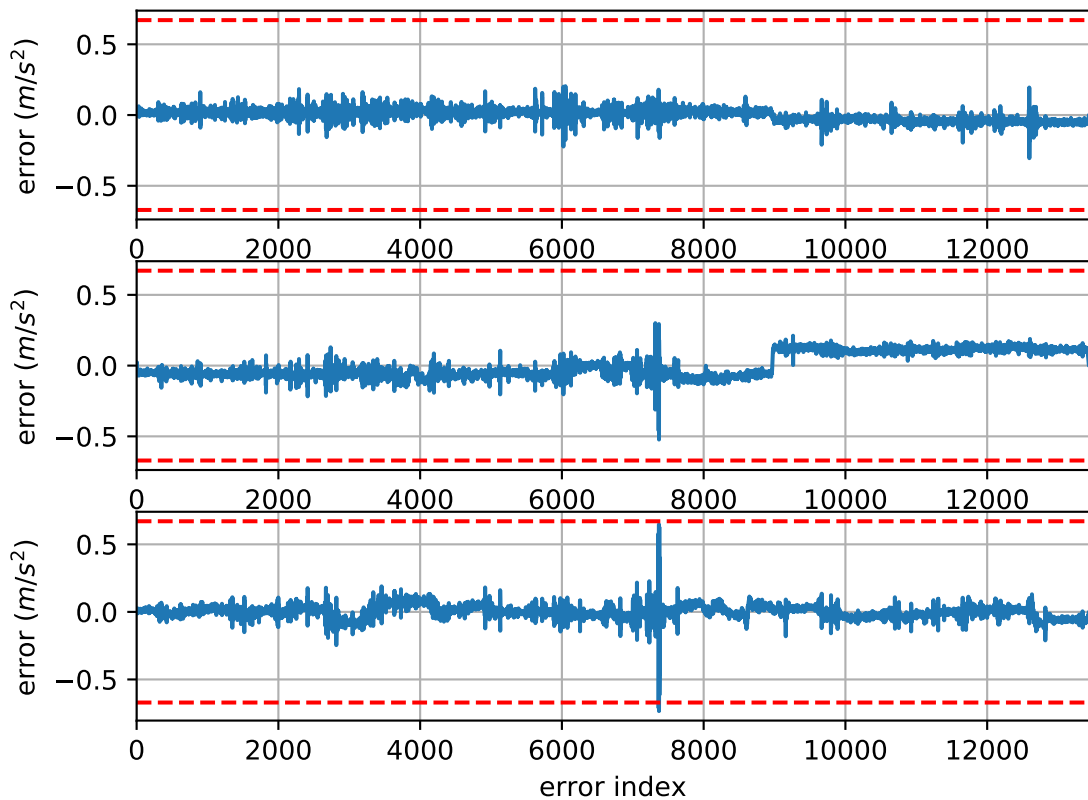
imu0: sample inertial rate



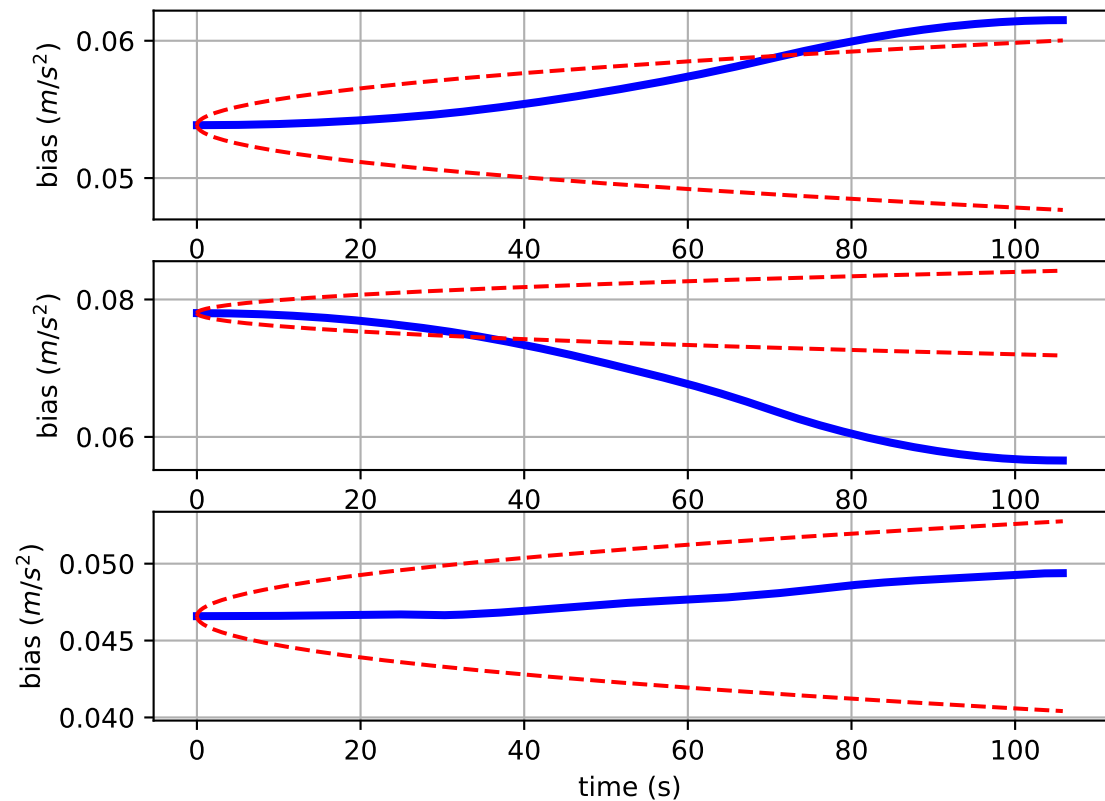
Comparison of predicted and measured specific force (imu0 frame)



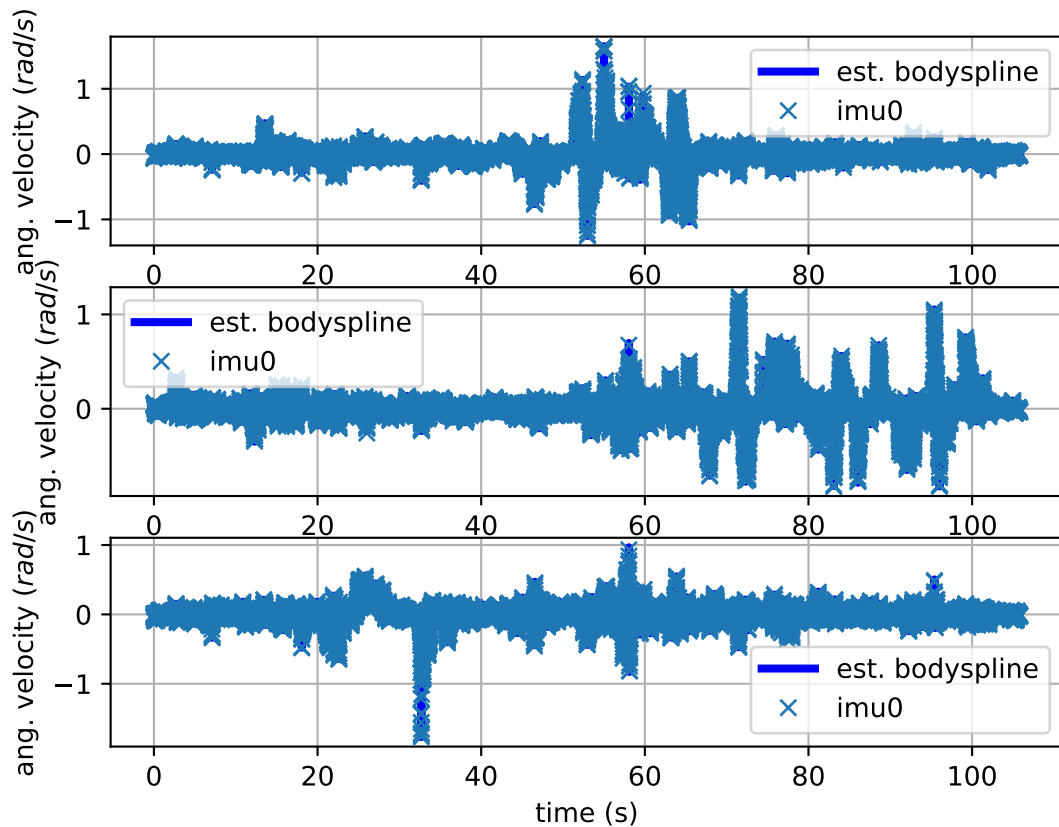
imu0: acceleration error



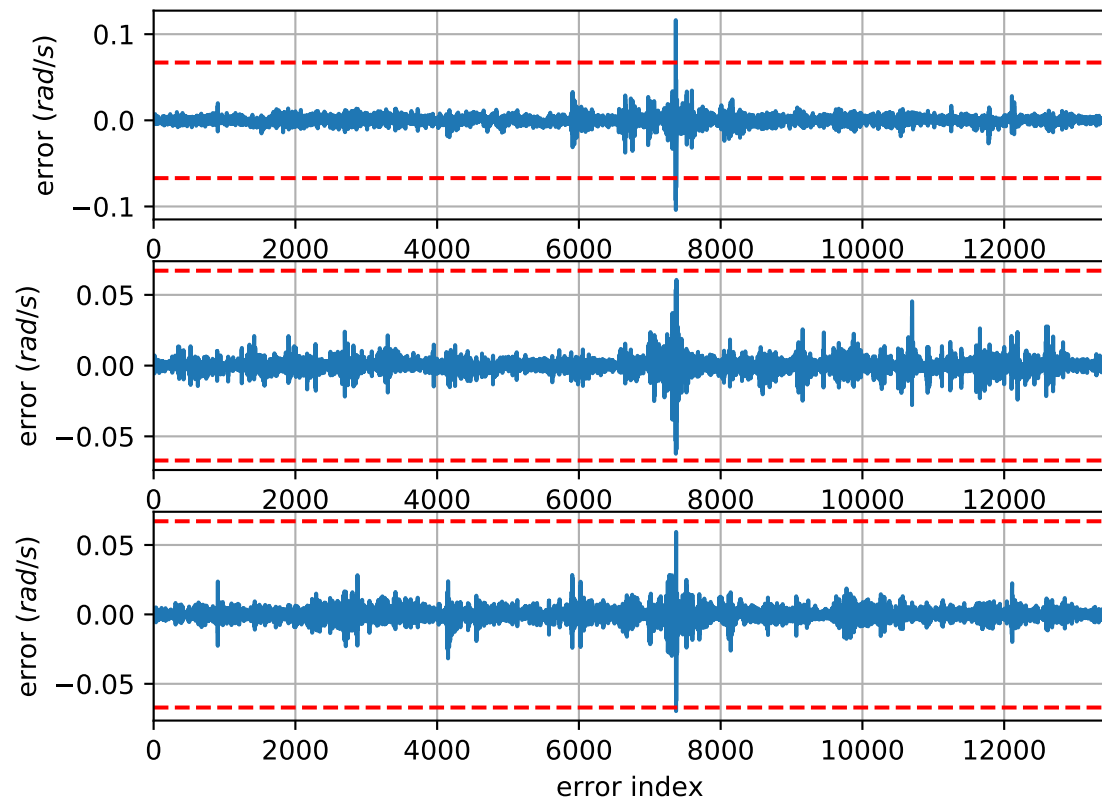
imu0: estimated accelerometer bias (imu frame)



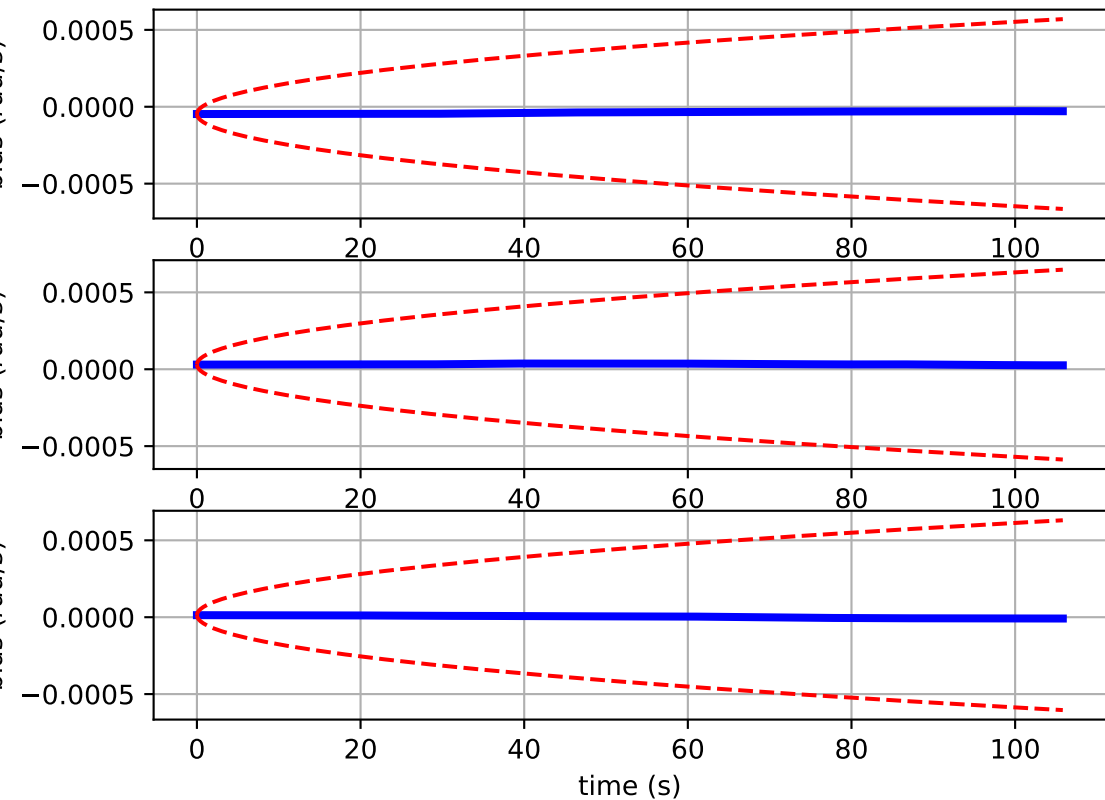
Comparison of predicted and measured angular velocities (body frame)



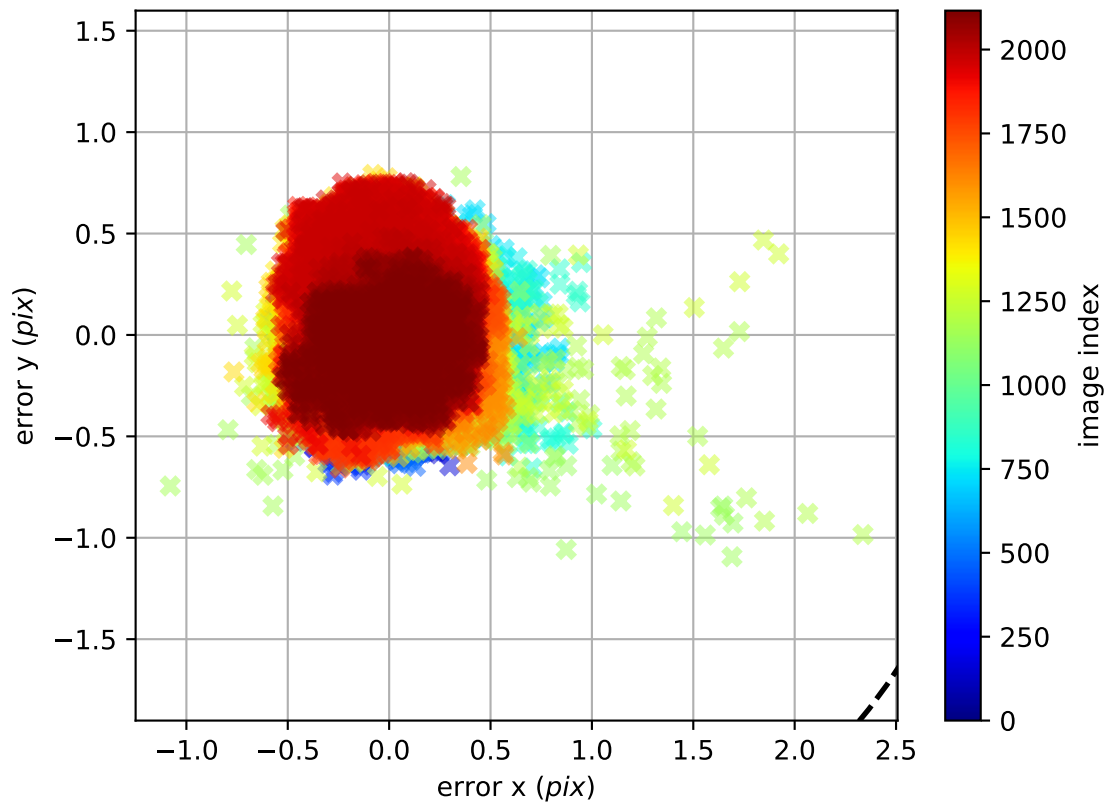
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

